

Ahmed Yassine Hachem

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Experience

AI Data Labeling Intern	<i>June 2024 – July 2024</i>
<i>PURA solutions</i>	<i>Sousse, Tunisia</i>
- Led an AI training initiative, labeling 20,000+ product images with detailed metadata and designing a scalable pipeline that reduced project time by 67%. - Achieved 98% labeling accuracy across 10,000+ images and collaborated with the CTO on QA for 20,000 labels, improving model accuracy by 20%.	
UI/UX Designer Intern	<i>April 2024 – June 2024</i>
<i>PURA solutions</i>	<i>Sousse, Tunisia</i>
- Redesigned the core app experience in Figma, modernizing the logo, homepage, and interface to enhance overall user engagement. - Optimized navigation and visual hierarchy through intuitive components, improved color schemes, and refined typography to make key features easier to discover.	
AI Engineer Intern	<i>June 2025 – July 2025</i>
<i>Brokins FR</i>	
- Collaborated with a co-worker to build two production ready AI tools from scratch for Brokins, a French insurance company. - Designed and implemented a RAG chatbot capable of searching across data from numerous French insurance companies and answering instantly - Built a smart insurance comparator that takes user preferences and recommends the most suitable insurer using AI-based matching - Handled personally the server deployment, connecting through SSH and managing setup with PUTTY and FileZilla, making sure both tools ran smoothly on Brokins infrastructure	

Skills

Software & Tools: *Figma, Adobe After Effects, Microsoft Excel, Adobe Premiere Pro, Adobe Photoshop, CapCut, VS code, Eclipse IDE*

Programming Languages: *Python, C, SQL, Java, HTML/CSS, Javascript*

Technologies: *Pandas, NumPy, Plotly, Seaborn, Matplotlib, scikit-learn, TensorFlow, Keras, Flask, RAG, Agent System, Multi-Agent System.*

Languages: *English (C1), French (B2), Arabic (native), German (A1), Spanish (A1).*

Projects

FertiFy <i>TypeScript, React, Next.js, Flask, Python, RAG.</i>
- Built FertiFy, a full-stack plant care web app using React, Next.js, Flask, Python, and RAG to provide interactive tools and AI-driven fertilizer and diagnosis recommendations. - Implemented two RAG-powered AI models (Fertilizer Predictor and Plant Diagnosis) and a Flask backend to process data and deliver accurate, real-time insights through a responsive UI. - Designed the app so users can log their plants, receive tailored care suggestions, and quickly diagnose issues, turning complex plant data into simple, actionable guidance.
NexusBI <i>TypeScript, React, Next.js, Flask, Python, RAG, Multi-Agents System.</i>
- Built NexusBI, a full-stack business intelligence platform using React, Next.js, Flask, Python, RAG, and a multi-agent system to turn raw data into actionable insights through interactive visualizations. - Engineered a data pipeline and Flask backend that ingest CSV/Excel/JSON files, preprocess complex datasets, and call Google Generative AI to generate comprehensive BI reports in natural language. - Designed specialized AI agents for data import, dataset management, and analysis, enabling automated end-to-end workflows from raw data ingestion to decision-ready recommendations with minimal user intervention.
Queue Intelligence System <i>Python, YOLOv8, OpenCV, Streamlit, CUDA, Agent System, Computer Vision.</i>
- Built an end-to-end queue intelligence system using YOLOv8, OpenCV, and Python to track individual customers in real time, compute dwell times, and export analytics-ready CSV/JSON outputs. - Designed polygon-based worker exclusion zones and disappearance logic so staff are detected but excluded from dwell and queue metrics, improving customer-only insights and reducing false positives. - Implemented a confusion-detection module that flags ID switches, occlusions, and re-entries, then calls an LLM (Google Gemini) to review video segments and automatically correct tracking errors with structured, high-confidence recommendations.

Achievements

HACKATHONS

Participated in 10+ hackathons over the past two years, with several notable achievements including the following:	
- <i>Hackathon 2.0 By AISEC</i>	<i>November 2023</i>
Second Place Hackathon Winner: Collaborated in a four-member team to develop an innovative IoT-driven app for water conservation, aligning with the UN Sustainable Development Goal for Clean Water and Sanitation. The solution integrated smart sensors and real-time data visualization to empower users with insights on household water consumption, driving sustainable water management practices.	
- <i>Coding moon challenge by Microsoft Club</i>	<i>February 2025</i>
Second Place Hackathon Winner: Collaborated in a team of 2 to develop a full-stack web application that visualizes data insights, integrating front-end technologies for user interaction and back-end services for data processing and analysis. Leveraged Google’s Gemini AI with Retrieval-Augmented Generation (RAG) to enhance data analysis capabilities, enabling the application to generate structured insights and comprehensive reports based on complex datasets.	
- <i>MutualHack by MAE insurance</i>	<i>May 2025</i>
Chosen as one of 103 participants from over 1,400 applicants, following a competitive preselection process based on my résumé and LinkedIn profile. Worked collaboratively in a small team to address real-time challenges in a fast-paced, high-pressure, competitive setting.	
- <i>ClimAdapt by AGEOS</i>	<i>October 2025</i>
<i>Second place Hackathon Winner:</i> Developed CO²LD an interactive environmental education game with a Climate Warning that combines AI, gaming, and sustainability to raise awareness about climate change	

Education

Pristini School of AI	Sousse, Tunisia
<i>applied artificial intelligence</i>	<i>Sep 2023 – May 2026</i>
Relevant courses: Deep learning Machine Learning Advanced Python programming C Java Data Engineering Data Administration Computer Vision Data Mining Big Data	